

# A Super-Horizon Vector Field as a Possible Origin of the Cosmic Dipole Anomaly

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## Abstract

We propose that the cosmic dipole anomaly (the  $> 5\sigma$  mismatch between the CMB dipole and the matter/galaxy/quasar dipole) may arise from a single super-horizon vector field with characteristic wavelength  $\lambda \gtrsim 3$ –10 times the observable-universe horizon. The CMB and matter dipoles would correspond to orthogonal components of this field. Galaxies could act as visible tracers aligning along the field lines. The model requires no new fundamental particles or exotic matter and is fully consistent with general relativity on sub-horizon scales. It makes immediate, quantitative predictions for Euclid, LSST, SKA, and SPHEREx that can be tested within the next 3–5 years.

**Disclaimer:** This is a phenomenological proposal based on current observational tensions in cosmology. It is intended as a possible mechanism to account for the cosmic dipole anomaly and offers one alternative interpretation for part of the observed Hubble flow. The ideas presented here are speculative and should be viewed as a hypothesis to be tested against forthcoming data rather than a definitive claim.

## 1 Introduction

The standard  $\Lambda$ CDM model faces several tensions in 2025–2026 data, the most statistically significant being the cosmic dipole anomaly. The temperature dipole in the cosmic microwave background and the number-count/redshift dipole in distant galaxies and quasars should agree in both direction and amplitude if the universe is homogeneous and isotropic on large scales. Instead, the matter dipole amplitude is observed to be 2–4 times larger than predicted from the CMB dipole, with the discrepancy now exceeding  $5\sigma$  in independent analyses from LOFAR, NVSS, RACS, and CatWISE (Secrest et al. 2025; LOFAR Collaboration 2025).

We propose a possible unified mechanism: a super-horizon vector field whose wavelength greatly exceeds the current observable horizon ( $\sim 93$  billion light-years).

## 2 The Super-Horizon Vector Field

The background metric is perturbed by a long-range vector potential  $A_\mu$  satisfying a Proca-like effective Lagrangian at leading order:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}m^2 A_\mu A^\mu + \frac{1}{M^2}(\partial_\mu A^\mu)(\partial_\nu A^\nu) + \dots,$$

where the effective mass  $m$  is extremely small ( $m \ll H_0$ ). The gradient of  $A_\mu$  couples differently to photons and to non-relativistic matter, naturally producing orthogonal projections that could manifest as the crossed CMB and matter dipoles.

The characteristic wavelength is estimated from the dipole excess amplitude ( $2\text{--}4\times$ ) and the requirement that the field remains coherent across the observable volume:

$$\lambda \gtrsim (3 - 10) \times (93 \text{ Gly}) \approx 280 - 930 \text{ Gly}.$$

Galaxies could act as visible tracers that align along the field lines, producing the observed cosmic web and large-scale filamentary structure. On the largest scales the field gradient offers a possible alternative interpretation for part of the observed Hubble flow as differential drift rather than uniform metric expansion.

A simple toy metric capturing the leading radial gradient is

$$ds^2 = -f(r) c^2 dt^2 + dr^2 + r^2 d\Omega^2,$$

where  $f(r)$  encodes the gentle super-horizon perturbation.

## 3 Consistency with Observations

The proposal is consistent with standard general relativity on solar-system and cluster scales (where the field would be screened or negligible) and does not conflict with existing sub-horizon data.

## 4 Falsifiable Predictions

The framework generates quantitative predictions testable with forthcoming surveys:

1. **Redshift dependence of the dipole excess:** The relative amplitude of the matter dipole should increase with redshift as  $A_{\text{matter}}(z)/A_{\text{CMB}} \propto (1+z)^\alpha$  with  $\alpha \approx 0.5\text{--}1.0$ . This will be measurable with Euclid (2027+) and LSST quasar samples.

2. **Filament and spin alignments:** Large-scale galaxy filaments and spins should show a statistically significant ( $\gtrsim 3\sigma$ ) preferred orientation correlated with the dipole axis (detectable in LSST and SKA data releases).

3. **Patch-dependent Hubble constant:** Local  $H_0$  measurements in different sky patches should vary by  $\Delta H_0/H_0 \sim 1\text{--}3\%$  correlated with position relative to the dipole axis (testable with future Cepheid and strong-lensing samples).

4. **Higher multipoles:** The quadrupole and octupole in future all-sky catalogs will exhibit non-random correlations with the dipole direction, revealing the harmonic structure of the super-horizon mode (SPHEREx and Euclid will provide the necessary sensitivity).

These signatures are specific, quantitative, and accessible within the next 3–5 years.

## 5 Discussion

This super-horizon vector field offers one possible, economical explanation for the cosmic dipole anomaly. It may also contribute to an alternative interpretation of part of the observed Hubble flow via differential drift through a large-scale gradient. The proposal is conceptually simple and makes concrete, near-term observational predictions.

## 6 Conclusion

A super-horizon vector field with wavelength  $\lambda \gtrsim 3$ –10 times the observable-universe horizon could naturally account for the cosmic dipole anomaly. The model is falsifiable with data expected from Euclid, LSST, SKA, and SPHEREx. We encourage the community to test these predictions in forthcoming large-scale surveys.

## Acknowledgments

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## References

- [1] Secrest et al. (2025). The Cosmic Dipole Anomaly. *Reviews of Modern Physics* (colloquium).
- [2] LOFAR Collaboration (2025). Large-scale anisotropy in radio galaxies.
- [3] Planck Collaboration (updated analyses, 2026).

### Open for Collaboration

Independent researchers are invited to test or extend these proposals. arXiv endorsement requests are welcome — please contact the author at [jorihuela@alum.mit.edu](mailto:jorihuela@alum.mit.edu).